Raw arrays to a defended result: the order is the content

1. Look at the data

569 rows x 30 columns
506 missing cells (2.96%)
212 malignant of 569

2. Split before anything else

398 train / 171 test
stratified on the label
the test rows are now sealed

3. Get a baseline first

always predict the majority
accuracy 0.6257
AUC 0.5000

then

4. Put every learned step in one pipeline

median imputer -> scaler
-> logistic regression
all three refitted per fold

5. Tune inside, score outside

5-fold grid inside
5-fold evaluation outside
nested AUC 0.9937 +/- 0.0086

6. Refit on all training rows

C = 0.1, penalty l2
chosen by the inner loop
not by the test set

then

7. Score the held-out rows exactly once

AUC 0.9966
accuracy 0.9766
and that is the number you report

8. Choose the threshold from the costs

10:1 for a missed cancer
misses 3 -> 1
false alarms 1 -> 6
the model did not change

9. Say how sure you are, and what would change it

171 test rows, so one patient
is worth 0.0058 of accuracy
the 95% interval spans 4.5 points